Jet Fuel Price Protection

Insurance-style caps vs traditional hedges for an airline, 2008–2026

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Summary

This report compares four ways an airline could have protected itself against rising jet fuel prices between 2008 and 2026: leaving its fuel unhedged, fixing the price with a swap, using a zero-cost collar, or buying a trigger–cap insurance policy. In each case 60% of the year's fuel is protected, and the protection is set up each December using only the information available at the time.

The swap removes the most risk. It reduced the uncertainty in the airline's monthly fuel bill by 84%, against 72% for the collar and 37% for the policy. The policy's figure barely improves when only budget overruns are counted (39%), which suggests that its 10% trigger and its cap, rather than the way risk is measured, are what limit its protection.

Where the policy stands out is cash. Swaps and collars have to be backed with collateral when prices fall, and in this backtest the peak swap collateral exceeded the policy's premium in 13 of 19 years; in 2020 it reached $82.0m against a premium of $6.2m. The policy was also cheaper than the swap in the four years in which prices fell after the hedge was set: 2014, 2015, 2020 and 2023.

Two further results matter for anyone pricing the product. Over the period the policy paid out $195.6m against $172.9m in premiums, so an insurer pricing on historical volatility would have needed a loading of at least 41% to break even. And the finding that the policy beats the swap only in a minority of years holds across every loading and trigger tested, at between 4 and 6 years out of 19.

Background

Fuel is an airline's largest cost it cannot control, and ticket prices cannot be raised fast enough to absorb a sudden spike. A finance director therefore chooses how much of next year's fuel to protect, and how.

Traditional protection comes from banks, as swaps or option structures such as collars. Both can be cheap or free upfront, but they are marked to market: when prices move against the airline, it must post cash collateral. An insurance-style policy works differently. The airline pays one premium, the policy pays out when prices rise above an agreed trigger, and there are no margin calls.

This project asks how these choices would have compared over the last two decades, in three terms a finance director cares about: what each cost, how much budget uncertainty each removed, and how much cash each tied up.

Data

The analysis uses daily US Gulf Coast kerosene-type jet fuel spot prices from FRED (series DJFUELUSGULF), in dollars per gallon. After cleaning there are 4,945 trading days from January 2007 to 14 October 2026, over which the price ranged from $0.41 a gallon on 27 April 2020 to $5.07 on 28 April 2022.

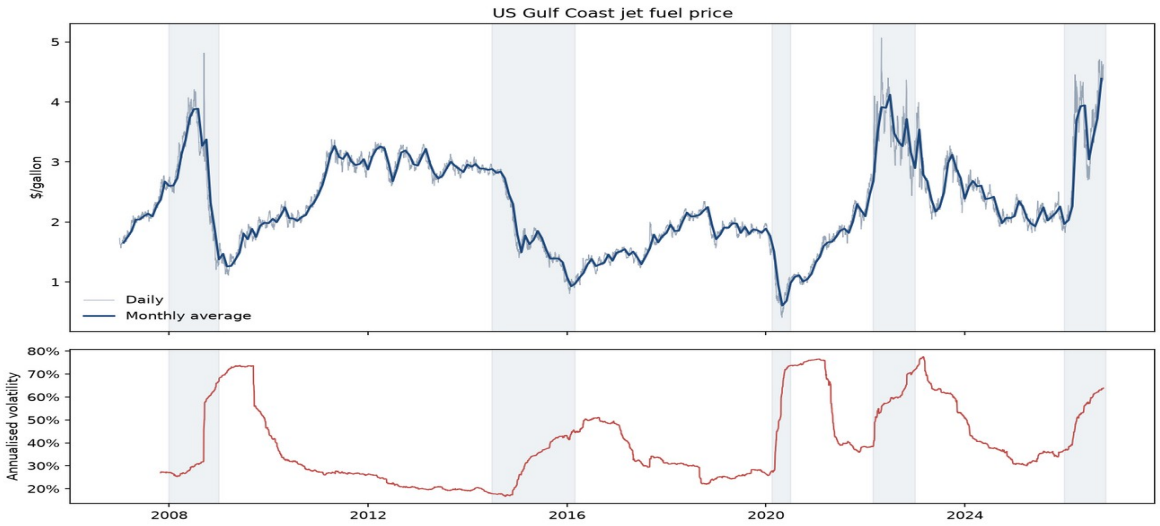


Figure 1. US Gulf Coast jet fuel price and rolling one-year volatility, 2007–2026. Shaded: 2008, the 2014–16 collapse, COVID, the Ukraine war and the 2026 spike.

The series is available from 2005, but 2005 and 2006 contain gaps of several weeks, so the analysis starts in January 2007. That still leaves a full year of history to estimate volatility before the first hedge programme in 2008. From 2007 onwards, 218 days are missing, almost all of them US market holidays. These are dropped rather than filled with the previous day's price, since forward-filling would create artificial days with no price change and understate volatility. Only 11 gaps of five days or more remain.

Fuel contracts typically settle on the average price over a calendar month, so daily prices are averaged by month for settlement. The final month is excluded when it is incomplete, which means 2026 covers January to September.

Volatility averaged 43.6% a year, but it moved a great deal: from under 20% in 2014 to over 75% after the 2020 crash and again in early 2023. As later sections show, those swings explain much of how the policy was priced.

Method

The backtest replays history one year at a time. Each year's hedge programme is set up on the last trading day of the previous year, using only the price and volatility known on that day, and it covers the following 12 calendar months. The swap fixes the price at the decision-day level. The policy's trigger and the collar's ceiling are both set 10% above that price, so the two products start protecting at the same point, and the policy's cap sits 50% above it. The collar's floor is then solved so that the call the airline buys is paid for exactly by the put it sells.

The policy is priced as a strip of twelve monthly call spreads using Black-76, with volatility measured over the previous 252 trading days, and a 25% loading is added to the fair value to reflect the insurer's margin. Each month then settles on its average price: the airline buys all its fuel at the market price, and the payoff on the protected 60% is netted against the bill. To capture the liquidity side, the swap and collar positions are also marked to market every day, and any negative value is treated as collateral the airline must post.

The monthly payoff per protected gallon is shown below, where P is the month's average price and F the decision-day price. A positive payoff is money received by the airline.

Strategy	Payoff per protected gallon
Swap	P − F
Zero-cost collar	max(P − ceiling, 0) − max(floor − P, 0)

Strategy	Payoff per protected gallon
Trigger–cap policy	min(max(P − trigger, 0), cap − trigger), after an upfront premium

Hedge effectiveness is 1 minus the ratio of a strategy's budget-surprise variance to the unhedged one. A downside version uses overruns only, counting months under budget as zero, since coming in under budget is not a risk the airline needs protecting from.

Cost and risk

The swap removed the most budget risk (84%), and the policy the least (37%), with the collar in between.

	Unhedged	Swap	Collar	Policy
Total cost, 2008–2026 ($m)	5,122	4,932	4,985	5,100
Average full year ($m)	267.7	261.5	263.9	267.9
Worst full year ($m)	404.1	369.6	366.4	375.6
Worst month ($m)	43.9	34.2	35.4	40.3
Typical monthly budget miss ($m)	6.05	2.42	3.23	4.81
Risk removed, all surprises	0%	84%	72%	37%
Risk removed, overruns only	0%	84%	72%	39%
Bad-month overrun, 95th percentile ($m)	12.96	5.19	6.54	8.55

The swap's 84% is a check on the model. With 60% of fuel fixed and 40% floating, its budget surprise is always 0.4 times the unhedged one, so theory predicts exactly $1 - 0.4^2 = 84\%$ on both measures.

The downside measure was added to test whether variance unfairly penalises the policy for letting the airline keep savings when prices fall. It barely moves the result (37% to 39%), so the policy's lower protection comes from its design. The 10% trigger leaves moderate rises uncovered, and the cap stops payouts in the most extreme months. In a bad month, the policy still cut the overrun by a third, from $13.0m to $8.6m; the swap cut it by 60%.

All three hedges cost less in total than going unhedged. This partly reflects the flat forward assumption (see Limitations) and should not be read as evidence that hedging saves money on average.

Performance year by year

The swap and collar win when prices rise after the hedge is set; the policy wins when they fall. The policy was cheaper than the swap in 4 of 19 years, cheaper than the collar in 5, and cheaper than going unhedged in 8.

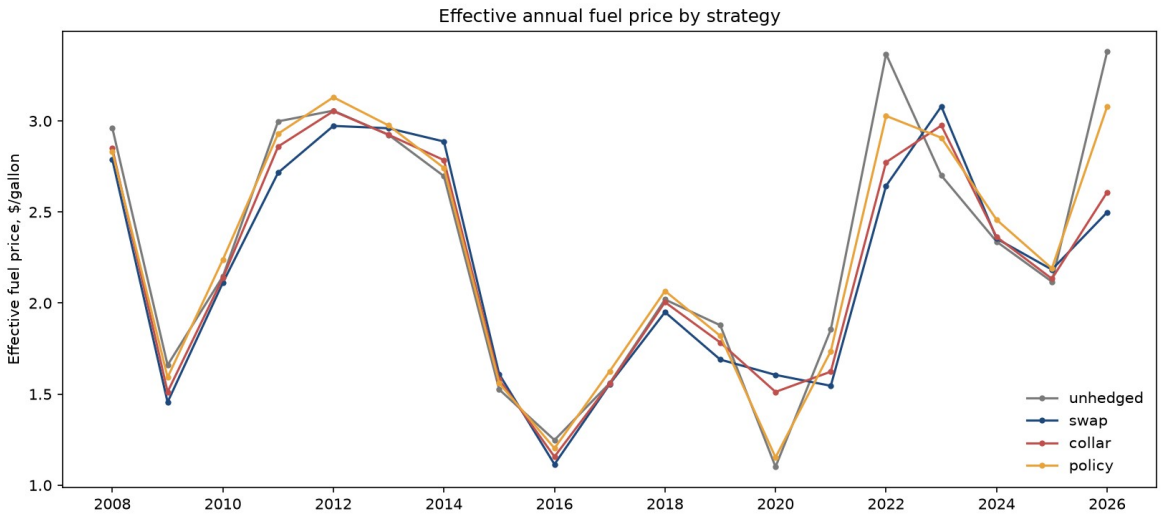


Figure 2. Effective annual fuel price by strategy, in dollars per gallon, 2008–2026. 2026 covers January to September.

The pattern is clearest in a handful of years. In 2008 all three hedges helped: the effective price fell from $2.96 a gallon unhedged to between $2.79 and $2.85. The crash came late in the year, by which point little of the programme set in December 2007 remained, so none of the hedges was badly caught out. The 2014–15 collapse was different. With prices falling from the level fixed at the start of the year, the swap became the most expensive choice, at $2.89 against $2.70 unhedged in 2014, while the policy stayed close to the market at $2.74.

2020 is the starkest case. The swap left the airline paying $1.61 a gallon when the market averaged $1.10, 46% more than if it had bought unhedged, whereas the policy cost $1.15, only its premium above the market. 2023 followed the same logic on a smaller scale. Hedges set at the high prices of late 2022 all turned out worse than buying unhedged at $2.70, but the policy ($2.91) did better than the collar ($2.98) and the swap ($3.08).

The spikes show the opposite side of the trade. In 2022 the swap brought the price down from $3.37 to $2.64, while the policy only reached $3.03, because prices spent most of the year above its cap. The current year looks similar so far, with the swap at $2.50 and the collar at $2.61, against $3.38 unhedged and $3.08 with the policy. In short, the four years in which the policy beat the swap (2014, 2015, 2020 and 2023) were all years in which prices fell after the decision day.

Cash and collateral

The policy’s clearest advantage is cash: peak swap collateral exceeded the policy’s upfront premium in 13 of 19 years, and the two largest calls came when airlines could least afford them.

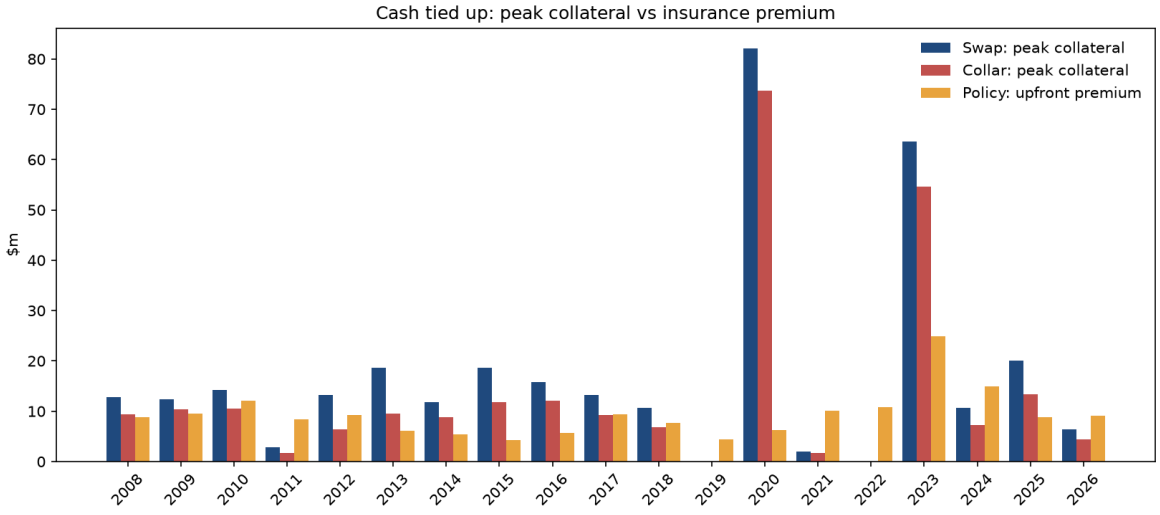


Figure 3. Peak collateral posted on the swap and collar each year, against the policy's upfront premium ($ millions).

The 2020 crash produced the largest calls. Prices collapsed at the same moment as airline revenues, and the swap required up to $82.0m in collateral and the collar $73.7m, against a policy premium of $6.2m. 2023 was the second-largest episode: hedges set at the high prices of late 2022 required up to $63.7m on the swap and $54.6m on the collar as prices drifted down, compared with a premium of $24.9m, the highest of the period. In years when prices rose steadily above the hedged level, as in 2019 and 2022, neither the swap nor the collar needed any collateral at all.

Collateral is not a cost: it is returned if prices recover. But it must be found in cash at the time it is called, often during the same downturn that cuts revenue. The policy's only cash flow is a premium known in advance. This is the main reason an airline might accept the policy's higher cost and lower risk reduction.

The insurer's side

At the modelled prices, the insurer would have lost money: the policy paid out $195.6m against $172.9m in premiums, leaving the airline $22.7m ahead despite a 25% loading. The break-even loading over Black-76 fair value was 41%.

The main cause is the volatility input. Premiums are priced on the past year's realised volatility, which lags the market in both directions. Going into 2020 and 2022, volatility was below its long-run average of 43.6%, so those years' policies were priced cheaply just before they were most needed. The reverse happened at the start of 2023: the premium was set straight after the most volatile stretch of the period, making it the most expensive of all ($24.9m), and prices then fell.

This should not be read as a sign that the product is underpriced in practice. A real insurer prices on implied volatility and its own actuarial models, not on a trailing one-year estimate. What the result does show is how strongly the pricing depends on the volatility assumption, and that a historical-volatility model on its own would not be a safe basis for setting premiums.

Sensitivity analysis

The main conclusion holds across every setting tested: the policy beat the swap in 4 to 6 of 19 years whatever the loading or trigger.

Loading	Trigger	Years policy beat swap	Overrun risk removed	Net payout to airline ($m)
0%	0%	5	52%	+85.3
0%	10%	5	42%	+57.3

Loading	Trigger	Years policy beat swap	Overrun risk removed	Net payout to airline ($m)
0%	20%	6	31%	+35.2
25%	0%	4	48%	+25.6
25%	10%	4	39%	+22.7
25%	20%	6	29%	+15.7
50%	0%	4	44%	−34.1
50%	10%	4	36%	−11.8
50%	20%	6	27%	−3.7

Of the two settings, the trigger has much more effect on protection. Raising it from 0% to 20% reduces the share of overrun risk removed by 17 to 21 percentage points, whatever the loading, whereas moving the loading from 0% to 50% reduces it by less than 10. The loading matters mainly for who comes out ahead: the airline gains at loadings of 0–25% and the insurer at 50%, which is consistent with the 41% break-even found above. A higher trigger also wins slightly more years against the swap, six rather than four or five, because its lower premium counts for most in years without a large price spike.

Limitations

The most important limitation is that every programme is set up on a single date, the last trading day of December. This keeps each year simple and comparable, but it makes the results depend on that one date. Airlines in practice layer their hedges 12 to 24 months ahead, so at any time they hold positions set at many different prices. The clearest consequence is 2008: hedges taken out in mid-2008, near the peak, would have generated large losses and collateral calls in the crash, but they do not appear here, which is why 2008 shows only modest collateral.

The other simplifications mainly affect the size of the numbers rather than their direction. Because free historical jet fuel forward curves are not available, the forward price for every month is taken to be the spot price on the decision day. In reality the swap rate would be the forward price, which already reflects expected price changes, so the swap's lower total cost partly reflects price drift within years and should not be read as evidence that hedging saves money. Premiums are based on historical rather than implied volatility, which, as shown above, makes them too cheap before spikes and too expensive after them. Plain Black-76 also slightly overprices contracts that settle on monthly averages, so the modelled premiums err on the high side.

Finally, the collateral figures assume daily margining with no threshold, no initial margin and no credit lines. They show the scale of the liquidity exposure rather than an exact cash requirement, and real credit arrangements would reduce them. The analysis also uses the US Gulf Coast benchmark because it is freely available, whereas European airlines buy against North West Europe prices, and it leaves out bank fees, credit charges and counterparty risk on both the bank and the insurance side.

Next steps

The natural next step is to remove the dependence on a single decision date. A first check would be to rerun the backtest with each of the twelve month-ends as the decision date and see whether the conclusions hold. A fuller version would model layered hedging, with a new 12-month programme set each month, as airlines actually hedge; this would capture the 2008 losses on hedges set near the peak.

Two further extensions would strengthen the analysis. The first is to compare a Brent cross-hedge with the jet fuel hedge, to measure basis risk, particularly in 2022, when refining margins pushed jet fuel well above crude. The second is to improve the pricing, replacing plain Black-76 with an average-price (Asian) approximation and testing a volatility estimate that responds faster than a one-year window.

Appendix: model settings

All results above use these defaults unless stated otherwise.

Setting	Value	Basis
Fuel bought per month	10 million gallons	Illustrative; results per gallon do not depend on it
Share of fuel protected	60%	Typical airline hedge ratio
Policy trigger	10% above decision-day price	Chosen; equal to the collar ceiling for a fair comparison
Collar ceiling	10% above decision-day price	Chosen
Collar floor	Solved for zero upfront cost	For example, 7.5% below the decision-day price at 40% volatility
Policy cap	50% above decision-day price	Chosen
Insurer loading	25% over Black-76 fair value	Assumption
Interest rate	3%	Simplification
Volatility	Trailing 252 trading days, annualised	Standard convention
Data	US Gulf Coast jet fuel spot, FRED series DJFUELUSGULF, daily	Free, long history